Active heads: velocity_forward

Auxiliary weight	velocity_forward	velocity_lateral	angular_velocity_y	track_curvature_0_50	track_curvature_1_00	track_curvature_1_50	track_curvature_2_00	track_curvature_2_50	track_curvature_3_00	track_curvature_3_50	track_curvature_4_00	track_curvature_4_50	track_curvature_5_00	track_curvature_5_50	track_curvature_6_00	track_curvature_6_50	track_curvature_7_00	track_curvature_7_50	track_curvature_8_00	wheel_slip_forward_front_left	wheel_slip_forward_front_right	wheel_slip_lateral_front_left	wheel_slip_lateral_front_right	wheel_slip_forward_back_left	wheel_slip_forward_back_right	wheel_slip_lateral_back_left	wheel_slip_lateral_back_right
0.1	1%	-5%	-7%	-5%	-6%	-6%	-10%	-4%	-2%	-1%	-0%	-1%	-1%	-1%	1%	-2%	0%	-2%	-0%	-19%	-18%	-9%	-7%	-19%	-17%	1%	1%
0.3	12%	-1%	-1%	-5%	-6%	-7%	-7%	-3%	-1%	-1%	-1%	-0%	-0%	-0%	0%	0%	2%	1%	2%	-3%	-1%	-3%	-3%	-3%	-1%	5%	5%
1.0	24%	6%	5%	-3%	-4%	-7%	**-11%**	-8%	-4%	-2%	-1%	-2%	-1%	1%	5%	-0%	3%	1%	1%	-4%	-4%	1%	1%	-5%	-5%	9%	8%

Active heads: velocity_lateral

Auxiliary weight	velocity_forward	velocity_lateral	angular_velocity_y	track_curvature_0_50	track_curvature_1_00	track_curvature_1_50	track_curvature_2_00	track_curvature_2_50	track_curvature_3_00	track_curvature_3_50	track_curvature_4_00	track_curvature_4_50	track_curvature_5_00	track_curvature_5_50	track_curvature_6_00	track_curvature_6_50	track_curvature_7_00	track_curvature_7_50	track_curvature_8_00	wheel_slip_forward_front_left	wheel_slip_forward_front_right	wheel_slip_lateral_front_left	wheel_slip_lateral_front_right	wheel_slip_forward_back_left	wheel_slip_forward_back_right	wheel_slip_lateral_back_left	wheel_slip_lateral_back_right
0.1	-7%	-0%	-1%	-2%	2%	-5%	-5%	-1%	3%	2%	0%	-0%	-0%	1%	1%	-0%	-0%	-2%	0%	-24%	-23%	-3%	-2%	-24%	-22%	6%	5%
0.3	-1%	3%	3%	-6%	-5%	-3%	-4%	2%	6%	7%	4%	2%	0%	1%	5%	1%	-1%	-2%	2%	-10%	-12%	-7%	-5%	-10%	-12%	11%	9%
1.0	2%	8%	8%	1%	-2%	-5%	**-14%**	-13%	-6%	-4%	-1%	-2%	-3%	-1%	4%	-3%	1%	-1%	-0%	-5%	-4%	1%	0%	-4%	-4%	**18%**	**15%**

Active heads: angular_velocity_y

Auxiliary weight	velocity_forward	velocity_lateral	angular_velocity_y	track_curvature_0_50	track_curvature_1_00	track_curvature_1_50	track_curvature_2_00	track_curvature_2_50	track_curvature_3_00	track_curvature_3_50	track_curvature_4_00	track_curvature_4_50	track_curvature_5_00	track_curvature_5_50	track_curvature_6_00	track_curvature_6_50	track_curvature_7_00	track_curvature_7_50	track_curvature_8_00	wheel_slip_forward_front_left	wheel_slip_forward_front_right	wheel_slip_lateral_front_left	wheel_slip_lateral_front_right	wheel_slip_forward_back_left	wheel_slip_forward_back_right	wheel_slip_lateral_back_left	wheel_slip_lateral_back_right
0.1	-1%	-2%	-3%	-4%	-8%	-9%	-9%	-4%	-0%	-1%	-0%	-1%	-0%	-1%	-3%	-2%	-2%	-1%	-1%	-22%	-16%	-7%	-4%	-21%	-15%	6%	5%
0.3	-6%	4%	6%	-4%	-4%	-5%	-8%	-3%	1%	2%	0%	-2%	-1%	-1%	1%	-2%	-2%	-2%	-0%	-13%	-11%	-2%	-1%	-13%	-11%	15%	13%
1.0	9%	11%	12%	3%	5%	0%	-4%	-1%	3%	2%	0%	0%	-1%	-1%	4%	-0%	3%	0%	-1%	-1%	-0%	5%	4%	-1%	0%	19%	17%

Active heads: curvature_close

Auxiliary weight	velocity_forward	velocity_lateral	angular_velocity_y	track_curvature_0_50	track_curvature_1_00	track_curvature_1_50	track_curvature_2_00	track_curvature_2_50	track_curvature_3_00	track_curvature_3_50	track_curvature_4_00	track_curvature_4_50	track_curvature_5_00	track_curvature_5_50	track_curvature_6_00	track_curvature_6_50	track_curvature_7_00	track_curvature_7_50	track_curvature_8_00	wheel_slip_forward_front_left	wheel_slip_forward_front_right	wheel_slip_lateral_front_left	wheel_slip_lateral_front_right	wheel_slip_forward_back_left	wheel_slip_forward_back_right	wheel_slip_lateral_back_left	wheel_slip_lateral_back_right
0.1	1%	3%	2%	26%	**30%**	**27%**	**18%**	**18%**	**16%**	9%	3%	-0%	-1%	-1%	2%	-2%	-1%	-2%	0%	-9%	-8%	-1%	-2%	-9%	-8%	3%	2%
0.3	7%	5%	3%	**48%**	**49%**	**50%**	**40%**	**41%**	**37%**	**22%**	6%	-1%	-1%	-2%	0%	**-5%**	-5%	-5%	-2%	-12%	-11%	-3%	-3%	-11%	-11%	8%	7%
1.0	1%	6%	5%	**54%**	**56%**	**58%**	**52%**	**52%**	**51%**	**27%**	6%	-3%	-5%	-6%	-4%	**-11%**	-7%	-7%	-5%	**-24%**	-20%	-3%	-3%	**-23%**	-19%	10%	9%

Active heads: curvature_medium

Auxiliary weight	velocity_forward	velocity_lateral	angular_velocity_y	track_curvature_0_50	track_curvature_1_00	track_curvature_1_50	track_curvature_2_00	track_curvature_2_50	track_curvature_3_00	track_curvature_3_50	track_curvature_4_00	track_curvature_4_50	track_curvature_5_00	track_curvature_5_50	track_curvature_6_00	track_curvature_6_50	track_curvature_7_00	track_curvature_7_50	track_curvature_8_00	wheel_slip_forward_front_left	wheel_slip_forward_front_right	wheel_slip_lateral_front_left	wheel_slip_lateral_front_right	wheel_slip_forward_back_left	wheel_slip_forward_back_right	wheel_slip_lateral_back_left	wheel_slip_lateral_back_right
0.1	3%	2%	2%	-1%	2%	1%	-3%	-1%	4%	8%	7%	**5%**	3%	1%	2%	-0%	-1%	0%	2%	-7%	-4%	-5%	-4%	-7%	-4%	8%	7%
0.3	-0%	-1%	-1%	0%	1%	3%	4%	5%	11%	17%	14%	**8%**	**7%**	8%	11%	2%	4%	3%	2%	-12%	-10%	-3%	-3%	-12%	-10%	4%	3%
1.0	3%	-2%	-5%	-4%	-1%	0%	6%	**17%**	**29%**	**37%**	**34%**	**20%**	**12%**	**10%**	12%	4%	3%	3%	4%	-14%	-13%	**-8%**	**-7%**	-14%	-13%	-4%	-4%

Active heads: curvature_far

Auxiliary weight	velocity_forward	velocity_lateral	angular_velocity_y	track_curvature_0_50	track_curvature_1_00	track_curvature_1_50	track_curvature_2_00	track_curvature_2_50	track_curvature_3_00	track_curvature_3_50	track_curvature_4_00	track_curvature_4_50	track_curvature_5_00	track_curvature_5_50	track_curvature_6_00	track_curvature_6_50	track_curvature_7_00	track_curvature_7_50	track_curvature_8_00	wheel_slip_forward_front_left	wheel_slip_forward_front_right	wheel_slip_lateral_front_left	wheel_slip_lateral_front_right	wheel_slip_forward_back_left	wheel_slip_forward_back_right	wheel_slip_lateral_back_left	wheel_slip_lateral_back_right
0.1	-10%	-1%	-2%	-4%	-7%	-6%	**-8%**	-2%	5%	3%	1%	1%	0%	1%	3%	-1%	0%	-1%	-1%	-12%	-14%	-5%	-3%	-12%	-14%	-2%	-1%
0.3	-11%	-5%	-5%	-5%	-5%	-7%	-4%	4%	**9%**	6%	1%	1%	1%	1%	1%	1%	0%	0%	2%	-19%	-18%	-9%	-7%	-19%	-18%	-2%	-1%
1.0	-16%	-5%	-5%	-4%	**-10%**	-14%	**-11%**	-7%	-1%	2%	-2%	-1%	1%	2%	3%	-1%	-1%	-0%	2%	-24%	-27%	-9%	-6%	-24%	-28%	-4%	-3%

Active heads: wheel_slip_front_forward

Auxiliary weight	velocity_forward	velocity_lateral	angular_velocity_y	track_curvature_0_50	track_curvature_1_00	track_curvature_1_50	track_curvature_2_00	track_curvature_2_50	track_curvature_3_00	track_curvature_3_50	track_curvature_4_00	track_curvature_4_50	track_curvature_5_00	track_curvature_5_50	track_curvature_6_00	track_curvature_6_50	track_curvature_7_00	track_curvature_7_50	track_curvature_8_00	wheel_slip_forward_front_left	wheel_slip_forward_front_right	wheel_slip_lateral_front_left	wheel_slip_lateral_front_right	wheel_slip_forward_back_left	wheel_slip_forward_back_right	wheel_slip_lateral_back_left	wheel_slip_lateral_back_right
0.1	19%	5%	3%	4%	3%	1%	**-7%**	-10%	-6%	-2%	-1%	-1%	-2%	-3%	-2%	-2%	-0%	-1%	0%	**25%**	**33%**	0%	1%	**26%**	**34%**	8%	8%
0.3	22%	5%	3%	7%	3%	-8%	**-17%**	-12%	-5%	-2%	0%	-3%	**-5%**	-5%	-3%	-3%	-1%	-1%	0%	**26%**	**29%**	-6%	-3%	**27%**	**29%**	9%	8%
1.0	9%	-25%	-26%	4%	9%	-6%	**-21%**	-13%	-7%	-6%	-6%	**-9%**	**-8%**	-5%	-2%	-4%	-4%	-6%	-5%	6%	14%	-32%	-24%	7%	14%	-23%	-19%

Active heads: wheel_slip_front_lateral

Auxiliary weight	velocity_forward	velocity_lateral	angular_velocity_y	track_curvature_0_50	track_curvature_1_00	track_curvature_1_50	track_curvature_2_00	track_curvature_2_50	track_curvature_3_00	track_curvature_3_50	track_curvature_4_00	track_curvature_4_50	track_curvature_5_00	track_curvature_5_50	track_curvature_6_00	track_curvature_6_50	track_curvature_7_00	track_curvature_7_50	track_curvature_8_00	wheel_slip_forward_front_left	wheel_slip_forward_front_right	wheel_slip_lateral_front_left	wheel_slip_lateral_front_right	wheel_slip_forward_back_left	wheel_slip_forward_back_right	wheel_slip_lateral_back_left	wheel_slip_lateral_back_right
0.1	-2%	5%	4%	1%	-2%	-3%	-2%	3%	6%	6%	3%	2%	2%	3%	6%	0%	0%	1%	3%	-15%	-16%	5%	4%	-15%	-15%	5%	4%
0.3	1%	7%	6%	-1%	2%	-1%	-6%	-6%	-3%	-2%	-1%	-2%	-4%	-3%	1%	-2%	2%	-1%	1%	-8%	-7%	8%	8%	-8%	-7%	6%	5%
1.0	13%	8%	7%	7%	5%	-2%	-8%	-5%	-0%	1%	0%	1%	-0%	1%	4%	-0%	2%	0%	1%	-6%	-6%	**14%**	**12%**	-6%	-5%	10%	9%

Active heads: wheel_slip_rear_forward

Auxiliary weight	velocity_forward	velocity_lateral	angular_velocity_y	track_curvature_0_50	track_curvature_1_00	track_curvature_1_50	track_curvature_2_00	track_curvature_2_50	track_curvature_3_00	track_curvature_3_50	track_curvature_4_00	track_curvature_4_50	track_curvature_5_00	track_curvature_5_50	track_curvature_6_00	track_curvature_6_50	track_curvature_7_00	track_curvature_7_50	track_curvature_8_00	wheel_slip_forward_front_left	wheel_slip_forward_front_right	wheel_slip_lateral_front_left	wheel_slip_lateral_front_right	wheel_slip_forward_back_left	wheel_slip_forward_back_right	wheel_slip_lateral_back_left	wheel_slip_lateral_back_right
0.1	9%	7%	5%	5%	4%	-0%	-6%	-6%	-6%	-5%	-3%	-6%	-5%	-5%	-5%	-5%	-5%	-3%	-2%	15%	**23%**	0%	1%	17%	**24%**	11%	10%
0.3	23%	11%	9%	3%	7%	0%	-12%	-11%	-5%	-2%	-0%	-2%	-2%	-2%	-2%	-3%	2%	-1%	-0%	**35%**	**42%**	2%	2%	**36%**	**43%**	**17%**	**16%**
1.0	12%	-30%	-32%	2%	-2%	**-17%**	**-29%**	**-22%**	-12%	-8%	-8%	-8%	**-7%**	-7%	-3%	-6%	-3%	-5%	-3%	10%	16%	-40%	-29%	11%	16%	-23%	-19%

Active heads: wheel_slip_rear_lateral

Auxiliary weight	velocity_forward	velocity_lateral	angular_velocity_y	track_curvature_0_50	track_curvature_1_00	track_curvature_1_50	track_curvature_2_00	track_curvature_2_50	track_curvature_3_00	track_curvature_3_50	track_curvature_4_00	track_curvature_4_50	track_curvature_5_00	track_curvature_5_50	track_curvature_6_00	track_curvature_6_50	track_curvature_7_00	track_curvature_7_50	track_curvature_8_00	wheel_slip_forward_front_left	wheel_slip_forward_front_right	wheel_slip_lateral_front_left	wheel_slip_lateral_front_right	wheel_slip_forward_back_left	wheel_slip_forward_back_right	wheel_slip_lateral_back_left	wheel_slip_lateral_back_right
0.1	1%	1%	1%	-2%	-1%	-3%	-4%	-1%	2%	2%	2%	1%	1%	3%	4%	0%	3%	1%	2%	-15%	-14%	-3%	-3%	-15%	-14%	9%	8%
0.3	4%	5%	6%	-4%	-1%	-3%	-6%	-1%	3%	2%	1%	-1%	1%	3%	5%	1%	2%	-0%	1%	-16%	-16%	0%	-0%	-17%	-16%	19%	18%
1.0	-12%	5%	8%	-3%	-3%	-8%	-10%	-6%	-0%	-0%	-0%	-1%	-2%	-2%	3%	-2%	1%	-1%	0%	-9%	-4%	-3%	-1%	-9%	-4%	**26%**	**25%**

Active heads: all

Auxiliary weight	velocity_forward	velocity_lateral	angular_velocity_y	track_curvature_0_50	track_curvature_1_00	track_curvature_1_50	track_curvature_2_00	track_curvature_2_50	track_curvature_3_00	track_curvature_3_50	track_curvature_4_00	track_curvature_4_50	track_curvature_5_00	track_curvature_5_50	track_curvature_6_00	track_curvature_6_50	track_curvature_7_00	track_curvature_7_50	track_curvature_8_00	wheel_slip_forward_front_left	wheel_slip_forward_front_right	wheel_slip_lateral_front_left	wheel_slip_lateral_front_right	wheel_slip_forward_back_left	wheel_slip_forward_back_right	wheel_slip_lateral_back_left	wheel_slip_lateral_back_right
0.1	24%	**16%**	**16%**	**36%**	**38%**	**35%**	**24%**	**22%**	**20%**	**12%**	7%	**5%**	**5%**	6%	8%	2%	5%	2%	2%	**34%**	**41%**	**12%**	**10%**	**35%**	**42%**	**28%**	**25%**
0.3	29%	**17%**	**17%**	**40%**	**42%**	**39%**	**27%**	**28%**	**24%**	**16%**	**9%**	**6%**	4%	5%	8%	3%	1%	1%	-0%	**36%**	**43%**	**12%**	**10%**	**37%**	**43%**	**29%**	**27%**
1.0	13%	**15%**	**14%**	**41%**	**41%**	**36%**	**25%**	**24%**	**24%**	**16%**	7%	-1%	-1%	-0%	4%	-2%	-2%	-3%	-1%	11%	18%	**7%**	**5%**	12%	19%	**21%**	**20%**

velocity_forward
velocity_lateral
angular_velocity_y
track_curvature_0_50
track_curvature_1_00
track_curvature_1_50
track_curvature_2_00
track_curvature_2_50
track_curvature_3_00
track_curvature_3_50
track_curvature_4_00
track_curvature_4_50
track_curvature_5_00
track_curvature_5_50
track_curvature_6_00
track_curvature_6_50
track_curvature_7_00
track_curvature_7_50
track_curvature_8_00
wheel_slip_forward_front_left
wheel_slip_forward_front_right
wheel_slip_lateral_front_left
wheel_slip_lateral_front_right
wheel_slip_forward_back_left
wheel_slip_forward_back_right
wheel_slip_lateral_back_left
wheel_slip_lateral_back_right

Auxiliary target

Auxiliary weight

ΔR^2 vs. baseline (mean over runs)

0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6